

TIC 313352561

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 0.656091 d (ALIASED, see below)
-> corrected period: 1.312182 d
Epoch (t0) 3097.7736 BTJD
Duration 0.0840 d
Depth 2.734 % (27,340 ppm)
SNR 213.1

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

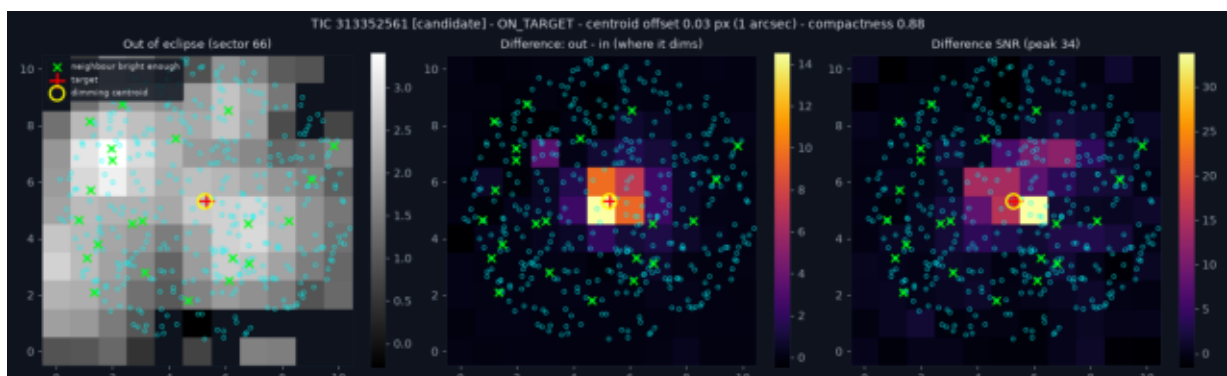
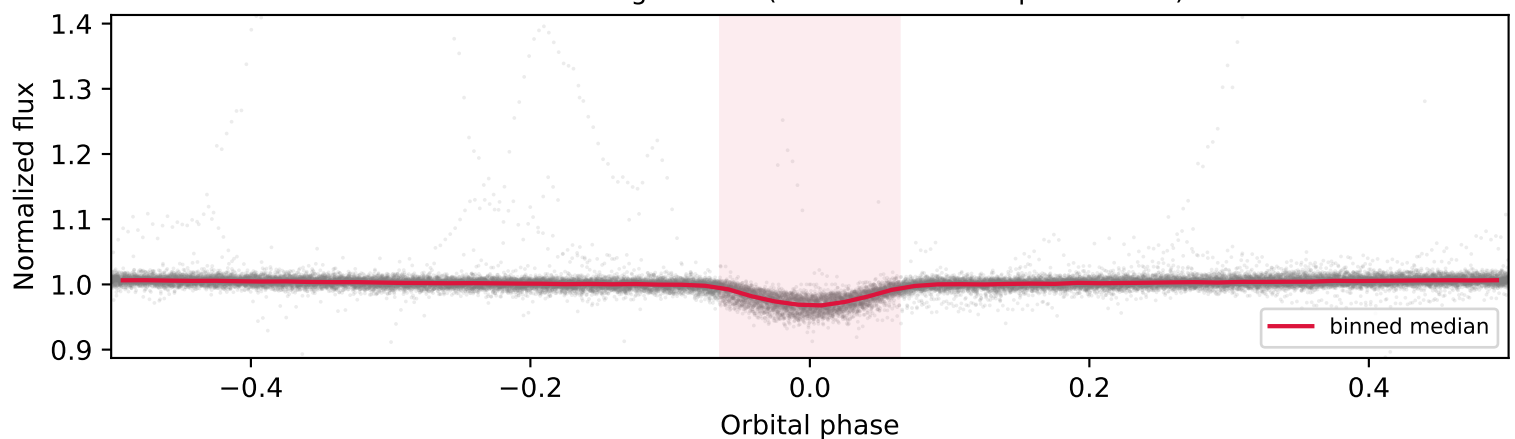
Eclipses observed 115 (gate ≥ 3)
Robust depth ratio 0.89 (gate ≥ 0.50)
Robust depth sigma 203.3 (gate ≥ 7.0)
Odd/even z-score 8.38 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 66 light-curve dip in this sector: 114.5 sigma t0 refined by -0.0100 phase
Difference-image peak SNR 33.89 (gate ≥ 5.0) Compactness 0.88 (gate ≥ 0.50)
Dimming centroid offset 0.03 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 313352547
1.70 px away, dTmag -0.40 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 8.38 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.312182 d, in any write-up.